Lesson 11: Wave After Wave

TIC-80

VIDEO GAME CREATION

This week: clearing a wave of enemies no longer means the game is over - it means the NEXT wave begins. Clear all 3, and that's when you really win.

Open your own [YourName]Game project from Lesson 10 - don't start a new one.

Tip: no new assets this week - multiple waves reuse your exact existing enemy spawn logic. This completes the Level 2 Complete milestone!

Setup

1. Launch TIC-80 and load your cart from Lesson 10.
2. Press Esc to reach the code editor.

Level 1 - Minimal (MVP)

Clearing a wave's enemies spawns a fresh wave, up to 3 total. Clearing wave 3 triggers the real win. The current wave number is shown on screen.

3. Add wave=1 and total_waves=3 near the top of your code.
4. Split your enemy-table setup out of start_game() into its own function spawn_wave() (code below) - just the table-populating loop, nothing else.
5. In start_game(), set wave=1 and call spawn_wave() at the end instead of populating the table directly.
6. At the bottom of TIC(), add a wave readout alongside your score/lives prints.
7. In the bullets loop's hit branch, revise the win-check into an if/else (code below): if this was the last wave, win as before; otherwise, advance the wave and call spawn_wave() again.
8. Press Esc, type run, press Enter, then play through - clearing wave 1 or 2 should spawn a fresh wave instead of ending the game; clearing wave 3 should show "You Win!".

```
function spawn_wave()
 enemies={}
 for i=0,4 do
  table.insert(enemies,{ex=20+i*20,ey=10,alive=true})
 end
end

function start_game()
 score=0
```

```
 lives=3
 wave=1
 enemies_remaining=5
 game_active=true
 bullets={}
 spawn_wave()
end

start_game()

-- bullets loop's hit branch (revised):
e.alive=false
enemies_remaining=enemies_remaining-1
score=score+10
if enemies_remaining<=0 then
 if wave>=total_waves then
  game_active=false
 else
  wave=wave+1
  enemies_remaining=5
  spawn_wave()
 end
end
```

Level 2 - Complete (Alpha/Beta)

Wave transitions are seamless and correct every time - no double-spawns, no skipped waves, exactly 3 waves before the real win, and restarting always returns to wave 1.

- Play a full 3-wave game: confirm waves 1 and 2 each spawn a fresh set of enemies with no win message, and only wave 3 triggers "You Win!".
- Let lives run out partway through wave 2 and confirm "Game Over" still works correctly, not a wave transition.
- Press R after a win or loss and confirm wave resets to 1, along with score and lives.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: briefly print a "Wave 2!" style message using a short countdown variable, distinct from the You Win!/Game Over prints.
- Structure: notice spawn_wave() is now called from two moments - start_game() and the bullets loop's else branch. The actual table-populating code only exists once.

If Something's Wrong

- Wave 1 immediately ends the game: confirm the if/else checks "wave>=total_waves" (not just enemies_remaining<=0 alone).
- New waves never spawn: confirm spawn_wave() is actually called inside the else branch, and that it replaces the enemies table with a fresh one (enemies={}) each time.
- Restarting leaves wave at whatever it was: confirm start_game() explicitly sets wave=1.

Want More?

Student Resource List > TIC-80 section, or tic80.com